1 Package guess

1.1 Introduction to package guess

Package `guess` guesses a formula for a sequence of numbers.

This package provides functions to find a formula for a sequence, of numbers within the hierarchy of expressions of the form, <rational function>, <product of rational function>, <product of product of rational function>, etc.

This package is a translation of the Mathematica package `Rate.m` by Christian Krattenthaler (Kratt at pap.univie.ac.at). The translation to Maple was done by Jean-Francois Beraud (Jean-Francois.Beraud at sic.sp2mi.univ-poitiers.fr) and Bruno Gauthier (Bruno.Gauthier at univ-mlv.fr).

All features of this package are due to C. Krattenthaler. The help text is due to Jean-Francois Beraud and Bruno Gauthier.

1.2 Definitions for package guess

`guess` [Function]

guess (*`l`*)
guess (*`l`*, *`nlevels`*)
guess (*`l`*, "*one*")
guess (*`l`*, "*nogamma*")

`guess(l)` tries to find a formula for a sequence within the hierarchy, of expressions of the form <rational function>, <product of rational function>, <product of product of rational function>, etc.

`guess(l, nlevels)`, where *nlevels* is a positive integer, does the same thing as `guess(l)`, but it searches only within the first *nlevels* levels.

`guess(l, "one")` does the same thing as `guess(l)` but it returns the first solution it finds.

`guess(l, "nogamma")` does the same thing as `guess(l)` but it returns expressions without gamma functions.

Examples:

```
guess([1,2,3]);
                                   [i0]


guess([1,4,9,16]);

                                      2
                                   [i0 ]
guess([1,2,6,24,120]);

                               i0 - 1
                               /===\
                                ! !
```

```
                          [ ! !   (i1 + 1)]
                            ! !
                           i1 = 1
```

guess(makelist(product(product(gamma(i)*i^2,i,1,j),j,1,k),k,1,8));

```
                      i0 - 1   i1 - 1    i2 - 1
                      /===\    /===\     /===\          2
                       ! !      ! !       ! !   (i3 + 3)
                     [ ! !   4  ! !   18  ! !   ---------]
                       ! !      ! !       ! !    i3 + 2
                      i1 = 1   i2 = 1    i3 = 1
```

guess([1,2,7,42,429,7436,218348,10850216]);

```
                    i0 - 1   i1 - 1
                    /===\    /===\
                     ! !      ! !   3 (3 i2 + 2) (3 i2 + 4)
                   [ ! !   2  ! !   -----------------------]
                     ! !      ! !   4 (2 i2 + 1) (2 i2 + 3)
                    i1 = 1   i2 = 1
```

guess(makelist(k^3+k^2,k,1,7));

```
Dependent equations eliminated:  (6)
                       i0 - 1
                       /===\
         2              ! !                      5040
      [i0  (i0 + 1), 2  ! !   (- --------------------------------------),
                        ! !        4        3         2
                       i1 = 1    i1  - 24 i1  + 245 i1  - 1422 i1 + 360

                                                       i0 - 1
                                                       /===\
                                                        ! !   (i1 + 1) (i1 + 2)
                                                      2 ! !   -----------------]
                                                        ! !            2
                                                       i1 = 1        i1
```

Note that the last example produces three solutions. The first and the last are equivalent, but the second is different! In this case,

guess(makelist(k^3+k^2,k,1,7),1);

or

guess(makelist(k^3+k^2,k,1,7),"one");

find only the solution `i0^2*(i0 + 1)`, which is a rational function, and is also the first function **guess** finds.

Appendix A Function and variable index

guess 2

(Index is nonexistent)